

Comparing Bilateral and Single-Sided Deaf Cochlear Implant Recipients in a Novel Speech-in-Noise and Localization Task

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Abstract

Objective. This study investigated how individuals who have undergone bilateral cochlear implantations (BiCIs) use compensatory head movements to optimize sound localization and speech-in-noise (SIN) performance compared with single-sided deaf users of cochlear implant (SSDCI) and normal hearing controls.

Study Design. Nonrandomized, prospective human-subject study.

Setting. Tertiary academic medical center.

Methods. Subjects were presented with Harvard IEEE sentences at varying signal-to-noise ratios (SNRs) in a darkened, semianechoic chamber. An electromagnetic head-tracking system monitored head movement. Outcomes included head movement patterns, localization accuracy, and SIN performance.

Results. BiCI localized targets less accurately with higher SNR-dependent variability than controls, but with greater accuracy than SSDCI. For SIN, SSDCI consistently showed superior performance to the BiCI, with no statistical significance found at any noise condition between SSDCI and controls. Across SNR, BiCI consistently initiated head movements more promptly compared to SSDCI, with controls predictably exhibiting the least delay. All CI recipients exhibited greater absolute displacement overall compared to controls. Although BiCI showed greater displacement compared to SSDCI, their pattern resembled controls, with decreased movement necessary as SNR became more favorable.

Conclusion. When comparing performance for localization and SIN across groups, the same impaired group does not perform most poorly at both binaural tasks; although SSDCI users maintain better speech understanding in noise, BiCIs tend to perform better on locating stimuli in space.

Preserved acoustic hearing in one ear does not provide a performance advantage across all tasks. The study underscores the intricate interplay of adaptive behaviors in CI users, showcasing successes and challenges in optimizing binaural hearing performance.

Keywords

bilateral sensorineural hearing loss, binaural hearing, cochlear implants, head movement, localization, single-sided deafness, speech-in-noise

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Binaural hearing is crucial for sound localization and comprehension of speech in noisy environments.¹ Individuals with deficits in binaural hearing are at significant safety risk as they navigate their daily environment due to their impaired localization ability. Additionally, maintaining a conversation in a group setting or in noisy environments becomes challenging, resulting in inevitable social isolation and a reduction in quality of life.^{2,3} Hearing rehabilitation aims to employ treatment modalities that attempt to restore binaural hearing. For that purpose, cochlear implants (CIs) have

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been utilized for individuals with bilateral sensorineural hearing loss (SNHL),⁴ and more recently for single-sided deafness (SSD).^{5,6}

Numerous studies have shown that bilateral cochlear implantation (BiCI) improves localization⁷⁻¹⁰ and speech-in-noise (SIN) understanding^{8,11} for individuals with bilateral SNHL with superior results to unilateral implantation for that population.¹²⁻¹⁴ Many other studies have investigated binaural performance in individuals with SSD with results showing improved ability,^{6,15,16} but significant variability stemming from diverse patient demographics and testing configurations challenges robust conclusions. Recently published work from our own lab further highlights the challenges in identifying and employing the most salient metrics for tests of binaural hearing.¹⁷

In prior investigation (Ibrahim et al, manuscript in preparation, 2024), we developed a novel testing paradigm to objectively assess sound-source localization and SIN performance while measuring head movement dynamics to gain insight into the compensatory mechanisms utilized by individuals utilizing CI for SSD. In this study, we evaluated the performance of BiCI users on this task and compared outcomes from the prior study. By contrasting the performance of individuals with electric stimulation in both ears (BiCI) to those with normal acoustic hearing in one ear and electric stimulation in the other (SSDCI), we hope to gain insight into the impact of mixed stimulation modality on binaural task performance.

Materials and Methods

Study Design

Conducted from September 2023 to June 2024 at a tertiary referral center, this prospective study compared localization accuracy, SIN performance, and head movement dynamics between BiCI, SSDCI, and normal hearing (NH) controls. The protocol was approved by the Institutional Review Board at the University of Michigan (HUM00190678).

Participants

Adults 18 years of age or older and fluent in English were included. Individuals with neurodegenerative disorders or retrocochlear pathology were excluded. CI recipients were recruited by phone, through social media advertisement, and from the otology clinic. NH participants were recruited using the University of Michigan Health Research Recruitment database (<https://umhealthresearch.org/>). Hearing status (HS) for NH subjects was verified using pure tone audiometry at a threshold of 25 dB HL across octave frequencies from 250 to 8000 Hz. For CI users, a minimum period of 6 months after device activation was required.

Testing Environment

Testing was conducted in a semianechoic chamber equipped with 24 loudspeakers arranged in 360° (spaced

15° apart) placed approximately 1.3 m from the participant at the level of the pinnae. Head movement was tracked using the Polhemus™ Fastrak® electromagnetic system (Polhemus,) that was affixed to the head with a custom-built band.

Stimulus and Procedure

Target stimuli were Harvard IEEE sentences presented at 70 dB SPL from one of 12 randomly selected loudspeakers. A brief, nontarget, orienting stimulus was added before each stimulus, consisting of the following sentence: “Hey, look over here. I am over here. Look over here.” Diffuse pink noise was used as a masker at seven predetermined signal-to-noise ratios (SNRs) (−10, −5, −2, 0, 2, 5, and 10 dB SNR). The total number of trials was 420, such that each loudspeaker and SNR combination was repeated five times. Custom software developed in MATLAB (R2019b; The Mathworks, Inc) was utilized to generate stimuli, record responses, register head position, and provide visual feedback regarding head position to the subjects by lighting LED lights along the array in real time. Live monitoring through a camera was done to ensure protocol compliance.

Participants began facing 0° azimuth, and they were instructed to turn towards the stimulus, indicate perceived source location with a button press, and repeat the target sentence. Head movement was not restricted, and real-time LED feedback controlled for potential effect of differential head, body, and eye movement. Subsequent trials were initiated by head position return to 0° azimuth. No changes were made to implant settings before participation.

Data Analysis

Localization accuracy was assessed using mean absolute error (MAE) and the average slope of the best fit regression line on the localization accuracy plot with TA on the *x*-axis and the response azimuth on the *y*-axis (see **Figure 1**). Robust linear regression was utilized when calculating the average slopes to reduce the effect of the outliers. A two-way analysis of variance (ANOVA) was used to analyze the effect of HS and SNR on performance.

Head movement dynamics were summarized as three salient components: onset delay, response time, and head displacement. *Onset delay* was calculated in seconds from the start of a trial until the participant moved their head past a minimum 15° threshold. *Response time* was defined as the duration from the start of a trial until the participants indicated their response. Trials consisting of stimuli presented from 0° azimuth were excluded for onset delay and response time due to errors resulting from the minimum onset criterion. Outliers exceeding three standard deviations were removed from onset delay and response time to address experimental errors and extreme values that deviate from the typical participant performance, often related to failed response capture. For the

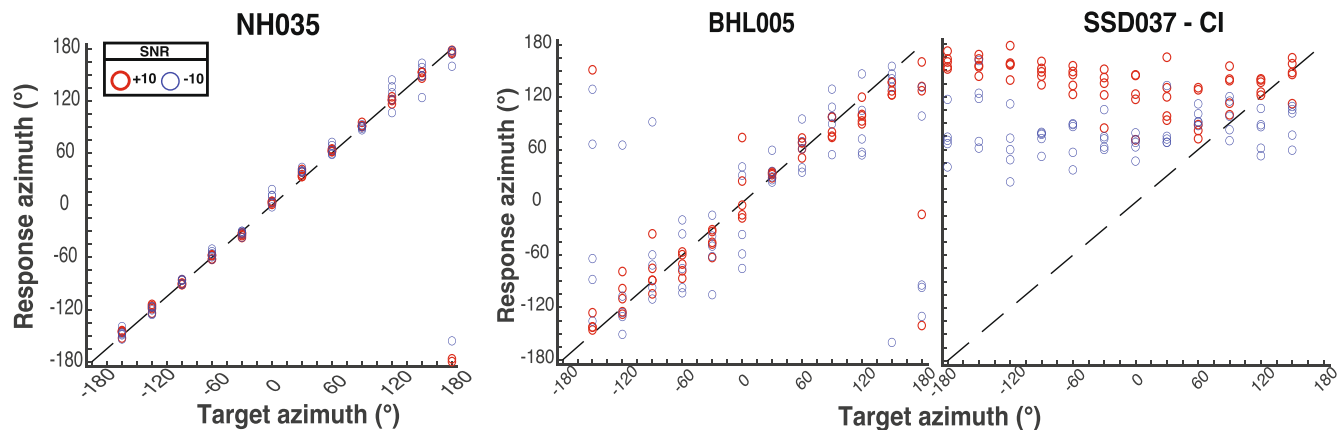


Figure 1. Example of individual localization performance from one subject in each group at +10 and −10 dB signal-to-noise ratio (SNR).

final component, *head displacement*, magnitude of movement regardless of the direction was considered.

Analyses for head movement dynamic subcategories were completed using three-way ANOVA assessing the effect of HS, SNR, and target azimuth (TA) on performance. When no significant interactions were detected, a comparison between BiCI and SSDCI was done using the nonparametric Wilcoxon rank-sum test as it does not require normal distribution assumption.

SIN performance was calculated as average percent correct per subject per SNR. Analysis using two-way ANOVA was used to characterize the effect to HS and SNR on performance.

Statistical analyses for all results were conducted using MATLAB, and all comparisons were assessed at the $\alpha = .05$ level. For multiple comparisons in all analyses, post hoc Bonferroni correction was done when significant interactions were detected.

Results

Subject Demographics

Twenty-eight NH subjects were included, 10 adults with CI for SSD, and 10 adults with BI for bilateral SNHL. All subjects met inclusion criteria, except for BHL012. Though their second-device activation was just under 6 months prior to testing, individual performance did not deviate from the group and was ultimately included. Detailed demographic information for hearing impaired subjects is summarized in **Tables 1** and **2**.

Localization Accuracy

Mean Absolute Error

The control group showed the lowest mean error (10.36°, SD = 5.21°) compared with BiCI who showed less accuracy than NH (40.51°, SD = 21.4°), but better overall accuracy than SSDCI (58.31°, SD = 19.78°) regardless of SNR. **Figure 2** shows average MAE by HS across SNR. Two-way ANOVA revealed a significant main effect of

HS ($F_{(2, 315)} = 413.79$, $P < .05$) and SNR ($F_{(6, 315)} = 10.57$, $P < .05$). A significant interaction of HS and SNR was observed ($F_{(12, 315)} = 2.44$, $P < .05$). Post hoc analysis showed differences in MAE between control and experimental groups across SNR.

A noteworthy example, the NH group at the least favorable −10 dB SNR, outperformed both SSDCI (M_{diff} = −35.12°, 95% CI [−52.66°, −17.59°], $P = 1.94e-10$) and BiCI (M_{diff} = −18.86°, 95% CI [−36.39°, −1.33°], $P < .05$) at the most optimal 10 dB SNR condition. Further, a comparison between SSDCI and BiCI at different SNR levels revealed multiple significant differences. Notably, when both groups were tested at the least favorable −10 dB SNR, the BiCI group showed better accuracy (M_{diff} = 22.7°, 95% CI [1.45°, 44.02°], $P < .05$). However, at the most favorable, 10 dB SNR, there was no statistically significant difference between the two groups (M_{diff} = 16.3°, 95% CI [−5.02°, 37.5°], $P = 1$). Detailed post hoc interactions are listed in Supplemental Appendix A, available online.

Slope (m)

NH subjects showed the highest slope values overall (0.94, SD = 0.0517), followed by BiCI (0.67, SD = 0.33) and SSDCI (0.46, SD = 0.39) regardless of SNR. Consistent with MAE, NH continued to show the best overall accuracy, followed by BiCI and SSDCI. **Figure 3** shows mean slope for all groups across SNR.

Results from two-way ANOVA showed a significant main effect of HS ($F_{(2, 315)} = 126.86$, $P < .05$) and SNR ($F_{(6, 315)} = 6.49$, $P < .05$). An interaction between HS and SNR was also found to be significant ($F_{(12, 315)} = 2.52$, $P < .05$). Post hoc test revealed that the SSDCI group at −10 dB SNR exhibited significantly worse performance compared to the BiCI group at all other tested SNRs. However, when comparing both groups specifically at −10 dB SNR, no significant difference was found (M_{diff} = −0.33, 95% CI [0.71, 0.04], $P = .23$). Although **Figure 3** illustrates a marked degradation in performance for BiCI recipients at all SNRs compared to control,

Table 1. Summary of Demographic Data for SSD CI Subjects

Subject ID	Age at testing	Gender	ADI national percentile	Etiology	Deaf ear	Duration of SSD	Age at CI (yrs)	Time since CI (mos)	Internal device	Sound Processor
SSD023	68	F	45	Sudden idiopathic	Left	16 mos	66	28	Mi250 Synchrony 2 Flex 28	Rondo 3
SSD024	62	F	18	Viral labyrinthitis	Left	9 mos	59	37	Nucleus C1632	Kanso 2
SSD025	61	F	10	Viral labyrinthitis	Right	14 mos	59	22	Nucleus C1612	N7
SSD031	78	M	39	Sudden idiopathic	Left	22 mos	71	84	Nucleus C1532	N8
SSD032	55	M	33	Viral labyrinthitis	Right	16 mos	54	6	Nucleus C1612	N8
SSD035	58	F	58	iatrogenic	Right	6 mos	57	12	Nucleus C1612	Kanso 2
SSD036	51	F	65	Sudden idiopathic	Left	7 mos	50	6	Nucleus C1632	N8
SSD037	49	F	42	Sudden idiopathic	Right	8 yrs	48	10	Nucleus C1632	N8
SSD038	71	M	67	Sudden idiopathic	Right	4 yrs	70	6	Nucleus C1612	N8
SSD039	60	M	Unknown	Sudden idiopathic	Right	3 yrs	59	12	Nucleus C1622	N8

significant differences between NH subjects and BiCI recipients in matched SNR conditions were observed only at -10 dB SNR ($M_{\text{diff}} = 0.51$, 95% CI [0.20, 0.82], $P = 4.27\text{e-}07$). In contrast, differences between NH and SSDCI were observed at all SNRs, notably at -10 dB SNR ($M_{\text{diff}} = 0.85$, 95% CI [0.54, 1.15], $P = 6.56\text{e-}19$) and 10 dB SNR ($M_{\text{diff}} = 0.36$, 95% CI [0.05, 0.66], $P < .05$).

Head Movement Dynamics

Figure 4 displays head movement onset delay, response time, and total absolute head displacement across SNR and TA.

Onset Delay

Irrespective of SNR and TA, the NH group showed the least onset delay (1.5 seconds, $SD = 0.55$ seconds), followed by BiCI (1.8 seconds, $SD = 0.91$ seconds) and SSDCI (2.6 seconds, $SD = 2$ seconds). Results from three-way ANOVA revealed a significant main effect of all three factors (SNR, HS, and TA) with a significant interaction between HS and SNR ($F_{(12, 3452)} = 2.09$, $P < .05$). Neither interactions between TA and other factors nor three-way interaction between SNR, HS, and TA were statistically significant ($P \geq .07$).

Examining the significant interaction between HS and SNR further through post hoc testing, multiple differences were noted; however, when SNR was matched, NH diverged from the BiCI group only at -5 and -2 dB SNRs ($P < .05$). When compared with NH at matched SNR conditions, SSDCI showed considerable statistically significant differences across SNR. Lastly, comparing SSDCI with BiCI at the same SNRs, a significant difference was observed across SNR ($P < .05$). The difference between SSDCI and BiCI was most pronounced at -5 dB SNR ($M_{\text{diff}} = 0.95$ seconds, 95% CI [0.44 seconds, 1.47 seconds], $P = 2.34\text{e-}09$). Detailed interactions are listed in Supplemental Appendix A, available online.

Response Time

The average response time independent of SNR and TA for the NH was 7.9 seconds ($SD = 3.43$ seconds); for the BiCI group, 8.49 seconds ($SD = 2.5$ seconds), and for the SSDCI group, 7 seconds ($SD = 2.78$ seconds). Three-way ANOVA considering HS, SNR, and TA as factors revealed a significant main effect of only HS ($F_{(2, 3465)} = 43.19$, $P < .05$). Comparison between SSDCI and BiCI showed a strong statistically significant difference between the groups ($Z = -10.92$, $P = 8.46\text{e-}28$) with the BiCI exhibiting longer response times.

Absolute Head Displacement

Generally, the NH group exhibited the lowest average displacement (116.03° , $SD = 45.7^\circ$) followed by SSDCI

Table 2. Summary of Demographic Data for Bilateral CI Subjects

Subject ID	Age at testing	Gender	ADI national percentile	Etiology	Age at HL onset	Age at CI		Internal device	
						Right	Left	Right	Left
BHL001	59	F	Unknown	Unknown	Gradual progressive	37	42	Nucleus CI24M	Nucleus CI24(RE)CA
BHL002	64	F	8	Hereditary	46	60	54	Nucleus CI632	Nucleus CI422
BHL003	18	F	32	Meningitis	18	18	18	Nucleus CI612	Nucleus CI612
BHL004	20	M	76	Auditory neuropathy	0	1	7	Nucleus CI24R (CS)	Nucleus CI512
BHL005	19	M	50	Usher syndrome, 1b	0	1	3	Nucleus CI24(RE)CA	Nucleus CI24(RE)CA
BHL007	24	F	83	Congenital	0	2	9, 13*	Nucleus CI24M	Nucleus CI24(RE)CA
BHL008	21	F	6	Connexin 26	0	5	1	Nucleus CI24(RE)CA	Nucleus CI24R (CS)
BHL010	57	F	29	Unknown	5	46	47	Nucleus CI24(RE)CA	Nucleus CI24(RE)CA
BHL011	33	M	57	Meningitis	4	21	9, 14*	Nucleus CI24(RE)CA	Nucleus CI24(RE)CA
BHL012	64	M	80	Noise exposure, radiation	40	64	61	Nucleus CI612	Nucleus CI612

*Reimplanted secondary to device failure.

(149.09°, SD = 83.8°) and BiCI (157.53°, SD = 75.43°). Further analysis using three-way ANOVA revealed a main effect of all three factors: HS ($F_{(2, 3780)} = 228.56$, $P = 1.97\text{e-}94$), SNR ($F_{(6, 3780)} = 9.065$, $P = 7.27\text{e-}10$), and TA ($F_{(11, 3780)} = 32.22$, $P = 4.75\text{e-}66$). Interaction between HS and SNR ($F_{(12, 3780)} = 3.7$, $P = 1.39\text{e-}05$) was also significant. **Figure 5** displays the effect of SNR on displacement across all groups. Interaction between HS and TA ($F_{(22, 3780)} = 13.13$, $P = 2.15\text{e-}46$) was found significant. The relationship between TA and displacement across the three groups is highlighted in **Figure 6**. No significant interaction was found between SNR and TA or in the three-way interaction ($P \geq .84$).

Bonferroni-corrected post hoc tests revealed that NH differed significantly from BiCI across SNR. Differences between NH and SSDCI were significant at all SNRs except at the challenging -10 dB, where the results did not reach statistical significance ($P = 1$). SSDCI and BiCI only differed at SNR -10 dB, with the BiCI group showing a higher degree of displacement ($M_{\text{diff}} = -39^\circ$, 95% CI $[-64.78^\circ, -13.93^\circ]$, $P = 6.62\text{e-}06$). Notably, when examining within-group performance, SSDCI showed no difference in head displacement across the different SNRs. In contrast, both NH and BiCI showed a statistically significant pattern of reduced degrees of displacement at optimal SNRs compared to more challenging conditions. Details of the interactions are available in Supplemental Appendix A, available online.

Examining interaction of HS and TA revealed a difference between NH and the two CI groups at most TAs, with the experimental groups exhibiting significantly higher degrees of displacement. SSDCI and BiCI only differed at TA 180°, with BiCI showing greater displacement ($M_{\text{diff}} = -39.83^\circ$, 95% CI $[-76.68^\circ, -2.98^\circ]$, $P < .05$).

SIN Performance

Figure 7 highlights the trend in the average SIN performance of the groups across SNR. The results from two-way ANOVA with HS and SNR as factors revealed a statistically significant main effect of both HS ($F_{(2, 315)} = 106.73$, $P = 4.07\text{e-}36$) and SNR ($F_{(6, 315)} = 274.09$, $P = 8.30\text{e-}122$) on SIN performance. A significant interaction was found between HS and SNR ($F_{(12, 315)} = 7.57$, $P = 2.6\text{e-}12$). Post hoc testing revealed multiple significant interactions across different SNR and HS combinations. However, when SNR was matched, results showed no difference between NH and SSDCI across any of the SNRs. A comparison between NH and BiCI showed differences in performance at SNRs starting with -2 dB ($P < .05$) onward. However, no significant difference was seen at -10 or -5 dB SNRs ($P = 1$). Similarly, comparing BiCI and SSDCI, differences were found starting at SNR 0 dB through 10 dB with no statistically significant difference in performance at the noisiest and most challenging conditions -2, -5, and -10 dB between the two groups.

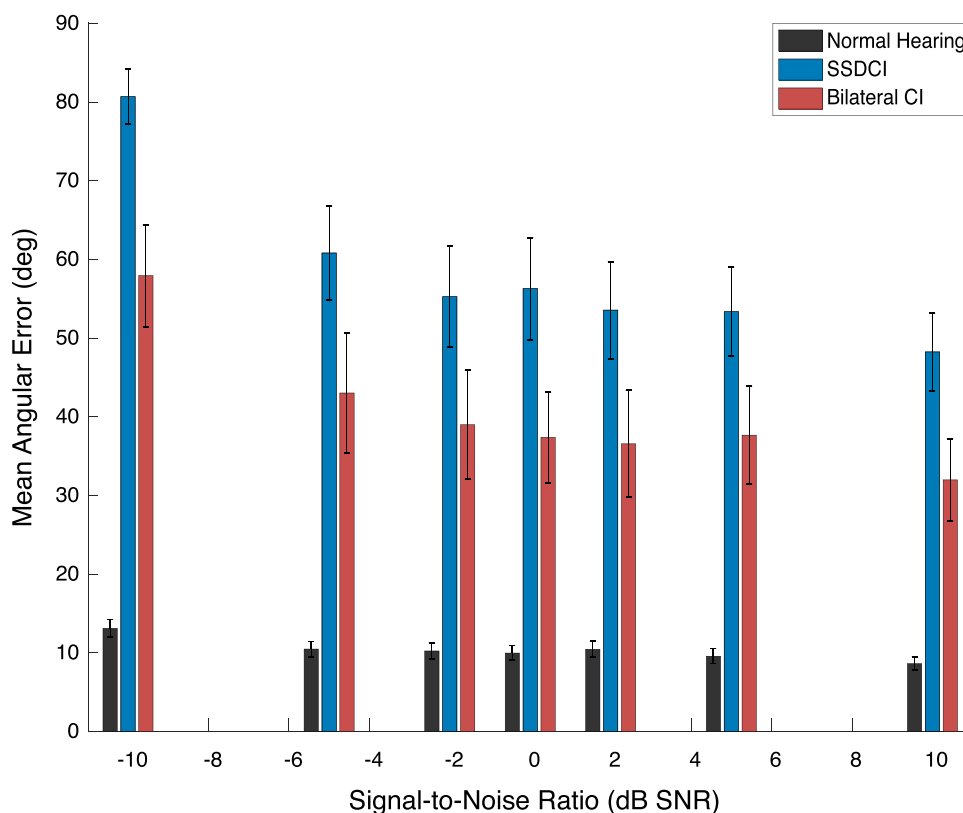


Figure 2. Average mean absolute error for the single-sided deaf users of cochlear implant (SSDCI) and bilateral cochlear implantation (BiCI) groups with the normal hearing group as reference across all signal-to-noise ratios (SNRs). Error bars represent the standard error of the mean.

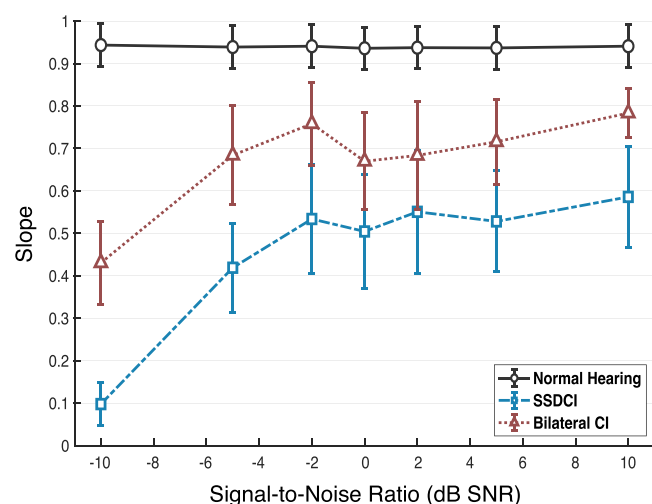


Figure 3. Mean slope values across signal-to-noise ratios (SNRs) for bilateral cochlear implantation (BiCI) (triangles), single-sided deaf users of cochlear implant (SSDCI) (squares), and normal hearing controls (circles). Error bars represent the standard error of the mean.

Discussion

Localization Accuracy

It has been widely described that sound-source localization accuracy is reduced in individuals with BiCI and SSDCI when compared with NH controls. Few prior

investigations have attempted to directly compare BiCI and SSDCI with the same protocol. Dorman et al¹⁸ compared these populations to controls using noise bursts as stimuli. They found that both BiCI and SSDCI groups exhibited a similar root-mean-square (RMS) error, which was significantly higher than that of controls. The same group also assessed localization using wideband noise and observed comparable RMS error between BiCI and SSDCI. Interestingly, the SSDCI group displayed high variability with a bimodal distribution.¹⁹

In employing a complex paradigm, our findings showed that CI recipients exhibit significantly poorer localization compared with controls at all noise settings. Notably, BiCI users exhibited enhanced performance over SSDCI users, evidence by higher slope values and lower MAE across various noise conditions. Statistical significance was observed only at the most extreme SNR for MAE. It is worth noting that although no statistical differences were observed in post hoc comparisons between NH and BiCI for slope except at -10 dB SNR, we acknowledge that this may reflect larger variance observed in the performance of CI recipients.

Both BiCI²⁰⁻²² and SSDCI²³ users are presumed to rely primarily on interaural level difference cues for sound localization, as the complex coincidence detector neuronal networks required for processing of timing cues are unlikely to be robustly activated by either place mismatched binaural electrical stimulation in BiCI or bimodal

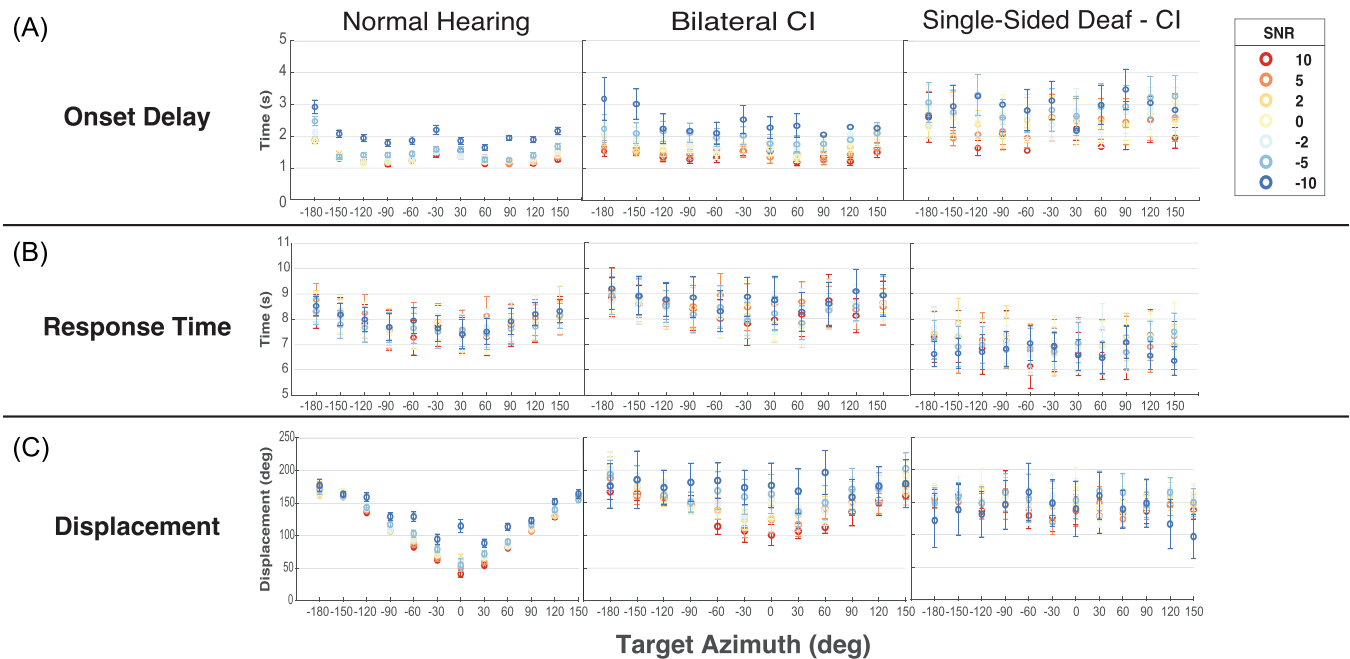


Figure 4. Head movement dynamics across signal-to-noise ratio and target azimuth for bilateral cochlear implantation (BiCI), single-sided deaf users of CI (SSDCI), and normal hearing controls. Onset delay is summarized in (A), response time in (B), and total absolute head displacement in (C).

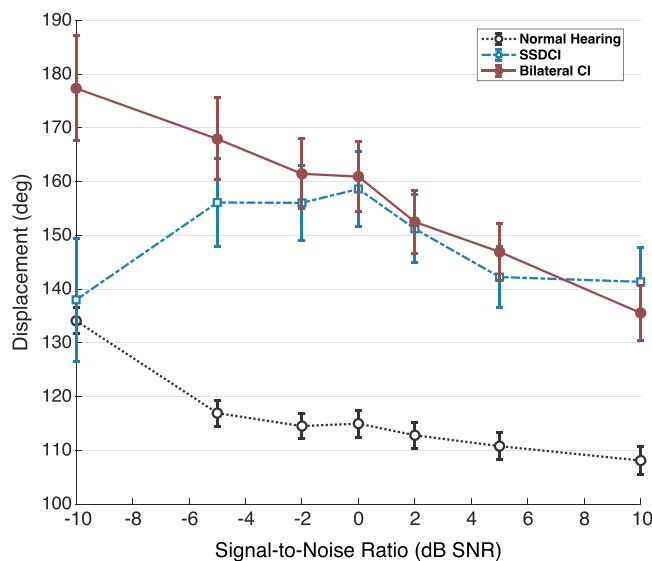


Figure 5. Total absolute head displacement trend for the three groups across signal-to-noise ratio (SNR). Error bars represent the standard error of the mean. SSDCI, single-sided deaf users of cochlear implant.

stimulation inherent with SSDCI.^{24,25} In addition to the place mismatch problem, interaural timing differences are not well replicated by the signal processing strategies currently in use; current clinic implant processors function independently and are not yet attempting to binaurally encode timing cues for bilateral CI users. Additionally, it is theorized that binaural deprivation experienced by those undergoing sequential bilateral implantation results in

degradation of neural timing sensitivity.^{21,26-28} Despite potentially suboptimal activation of binaural networks, BiCI users have the advantage of relatively more symmetrical auditory input compared with SSDCI users, who have one acoustic and one electric source.¹⁸ This discrepancy in auditory stimulation modality offers a potential explanation for the presented findings, underscoring the need for advancement in CI technology to enhance sound localization, particular by improving signal symmetry.

Head Movement Dynamics

Across all background noise levels, the BiCI group consistently initiated head movements more promptly compared to the SSDCI group. Controls predictably exhibited the least amount of onset delay; however, a statistically significant difference compared with the BiCI group was only observed at the least optimal SNRs. Furthermore, our CI recipients exhibited greater absolute displacement overall compared to controls. Interestingly, although the BiCI group showed greater displacement compared to SSDCI, their pattern resembled that of the NH group, with decreased amount of movement necessary as SNR became more favorable. Conversely, the SSDCI group displayed less head displacement at the least optimal SNRs (see **Figure 5**, SNR = -10 dB). Considering their reduced localization accuracy, this finding may reflect diminished engagement with the task, potentially due to the increased difficulty of accurately discerning sound sources. This behavioral adaptation suggests a threshold of auditory

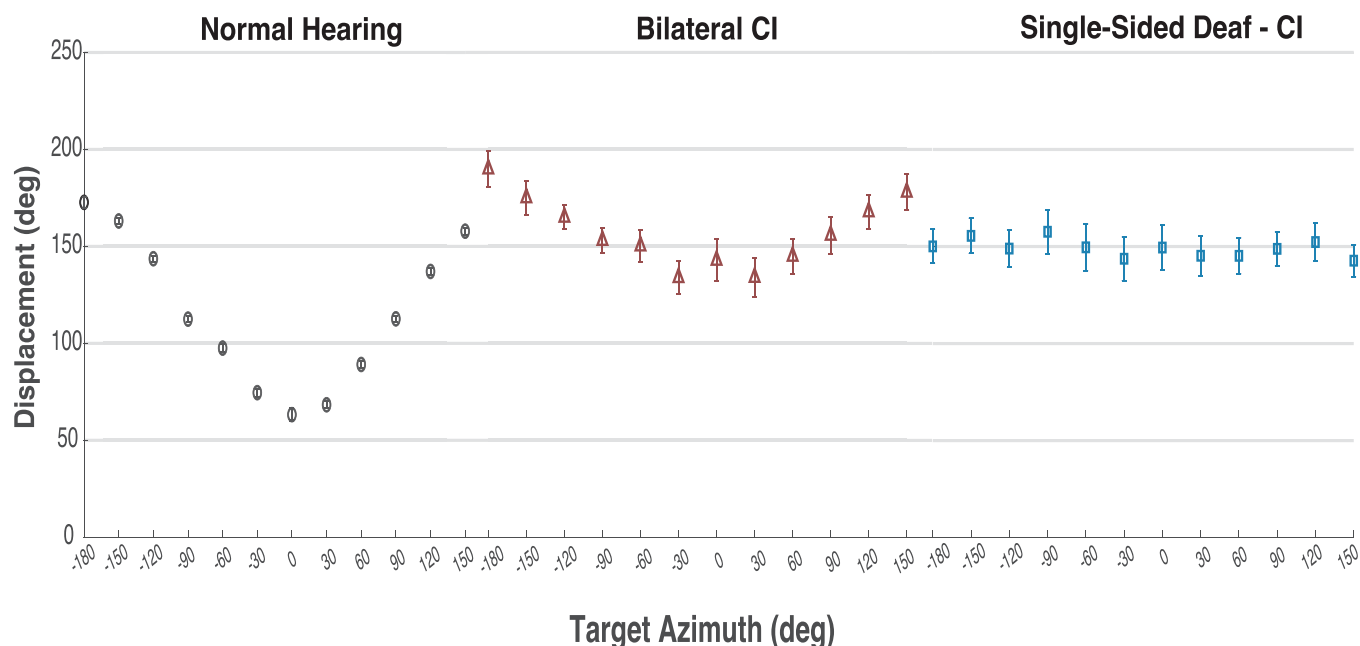


Figure 6. Pattern of total absolute head displacement in relation to target azimuth across the three groups. Error bars represent the standard error of the mean. CI, cochlear implant.

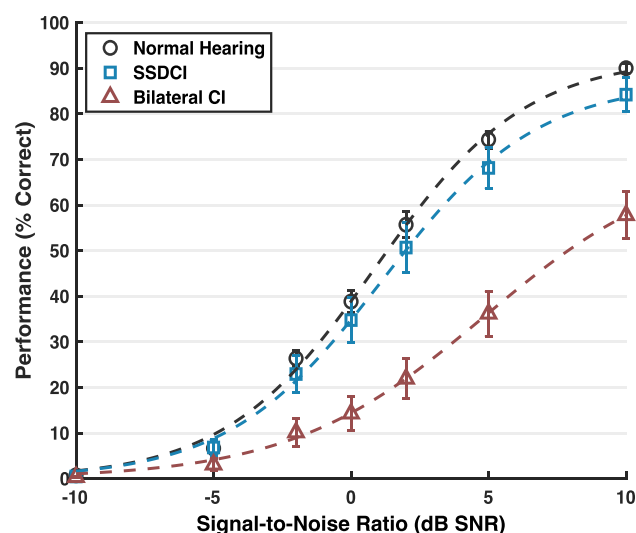


Figure 7. Speech-in-noise performance for the three groups across signal-to-noise ratio (SNR). Error bars represent the standard error of the mean. SSDCI, single-sided deaf users of cochlear implant.

challenge beyond which SSDCI users might limit their participation.

SIN Performance

Our SSDCI group performed similarly to controls, with no statistical significance found at any noise condition. The performance of all groups was impacted similarly at the noisiest SNR conditions. A clear significant pattern was evident at all other SNRs, with SSDCI consistently showing superior performance to the BiCI group. A

plausible explanation is that SSDCI individuals benefit from intact acoustic hearing in their unimpaired ear, utilizing temporal fine structure cues for speech perception.²⁹ In contrast, the BiCI group depends on implant technology alone, which does not offer equivalent temporal resolution, limiting speech comprehension capabilities, particularly under the most challenging conditions.

When comparing performance outcomes for the two binaural tasks (localization and speech understanding in noise) more generally across our groups, the most prominent finding appears to be that the same impaired group does not perform most poorly at both binaural tasks; although single-sided deaf CI users maintain better speech understanding in noise, bilateral recipients tend to perform better on locating stimuli in space. Preserved acoustic hearing in one ear does not provide a performance advantage across all tasks. Although perhaps not surprising for the proposed mechanistic explanations offered regarding stimulation mismatch, this finding may underscore an important target for future research into improvement of binaural hearing for individuals with CI. Namely, we should prioritize surgical techniques and programming strategies that give precedence to increasing stimulation symmetry between ears to the degree that is possible. Anatomy-based mapping strategies have been proposed as one method to achieve more equalized stimulation, but other novel programming strategies may prove to accomplish this end.

Conclusion

Cochlear implantation offers a vital solution for individuals with both bilateral hearing loss and SSD. This study

advances our understanding by uniquely assessing the integration of localization and SIN performance, more closely reflecting real-world auditory environments. It further delves into the head movement dynamics used by CI recipients as compensatory mechanisms during binaural tasks. The findings underscore the intricate interplay of adaptive behaviors in implant users, showcasing successes and challenges in optimizing binaural hearing performance.

Author Contributions

Obada Abdulrazzak, study concept and design participation, data acquisition, interpretation, and analysis, drafting of the manuscript, final approval of the manuscript; **Nadine I. Ibrahim**, data acquisition, review final approval of the manuscript; **Gerilyn Jones**, data acquisition, review and final approval of the manuscript; **Madison V. Epperson**, data acquisition, review and final approval of the manuscript; **Ilhan Onder**, study concept and acquisition software design, review and final approval of the manuscript; **Jackson Graves**, analysis software design, review and final approval of the manuscript; **Carolyn Kroger**, analysis software design, review and final approval of the manuscript; **Anahita H. Mehta**, analysis software design, interpretation, and analysis, drafting of the manuscript, final approval of the manuscript; **Renee M. Banakis Hartl**, study concept and design participation, data acquisition, interpretation, and analysis, drafting of the manuscript, final approval of the manuscript.

Disclosures

Competing interests: None.

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Supplemental Material

Additional supporting information is available in the online version of the article.

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